USER MANUAL · VERSION 1.0

IR GENERATOR

Capture your own speaker-cab impulse responses with the AXE I/O, for TONEX, AmpliTube and Fractal.



macOS 14+ · Apple silicon

Open source · MIT licence

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> **SCREENSHOTS** The screenshots in this manual come from the app's demo mode (`IRGenerator --demo`), which fills the library with simulated captures. Your own captures will look similar, but with the character of your cab, mic and room.

1 What IR Generator does

An **impulse response (IR)** is a short audio file that describes how something changes a signal, in this case how a guitar speaker, its cabinet, the microphone and its position colour the sound of an amp. Amp modellers such as TONEX, AmpliTube and the Fractal Axe-Fx, FM9 and FM3 use IRs in place of a real cab and mic.

IR Generator lets you make your own. It sends a test signal from the AXE I/O's **Amp Out** into a power amp and your cab, records the result through a mic, and works out the IR mathematically. You can then inspect it, compare it with other captures, combine mics, and export files in the exact formats your modellers expect.

Features at a glance

- **Accurate measurement.** Exponential sine sweep (the studio standard) or periodic pink/white noise, with averaging. Playback and recording run in the same audio callback, so they're locked sample for sample.
- **Four views.** Frequency response, impulse (waveform or energy-time), the raw recording, and a spectrogram. Overlay several captures to compare positions or mics.
- **Align & Blend.** Phase-align IRs from different mics to a fraction of a sample, then blend them without comb filtering.
- **Mic Swap.** Remove the capture mic's colouration for a speaker-only IR, or re-voice the capture as another mic.
- **One-click export.** Every length you need for Fractal and TONEX/AmpliTube in one go, with normalisation, fades, minimum phase and a high cut.
- **A safe library.** Each capture keeps a full-length master IR and the raw recording, so you can re-process or re-export at any time.

How a capture works

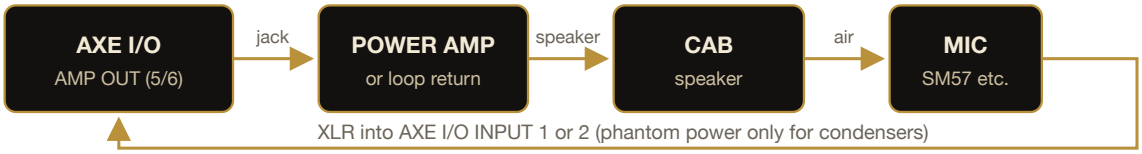
The default test signal is a **logarithmic sine sweep**: a tone that glides from 20 Hz to 20 kHz over a few seconds. The app records what the mic hears and divides the recording by the sweep in the frequency domain (deconvolution). What's left is the cab's impulse response. A useful property of the log sweep is that the power amp's harmonic distortion ends up *before* the main impulse in time, where the app trims it off, so the IR describes the cab and not the amp's distortion.

2 Equipment and wiring

You will need

- A Mac with Apple silicon running macOS 14 or later.
- An **IK Multimedia AXE I/O**. Its **Amp Out** (outputs 5/6) is designed to drive an amp input; the line outputs 1/2 and 3/4 work too, for a power amp with a line input.
- A **power amp** (or the power-amp section / effects-loop return of a valve amp) and the **cab** to capture.
- A **microphone** and an XLR cable. A Shure SM57 is the classic choice; any mic works.
- Optionally, a short instrument cable for the loopback latency calibration.

Signal chain



Set up the AXE I/O first

Use IK's control panel (or Axeio Silicon) before you capture:

- **To Amp Out: OFF.** When it's on, Input 1 is routed straight to Amp Out. With a mic on Input 1 that creates a feedback loop through the cab.
- **Direct monitoring down** for the mic input, for the same reason.
- **Phantom power** only if you're using a condenser mic. Dynamic mics and most ribbons don't need it, and some older ribbons can be damaged by it.
- Set the AXE I/O to **48 kHz**. The app does this for you when it opens the device.

Mic placement

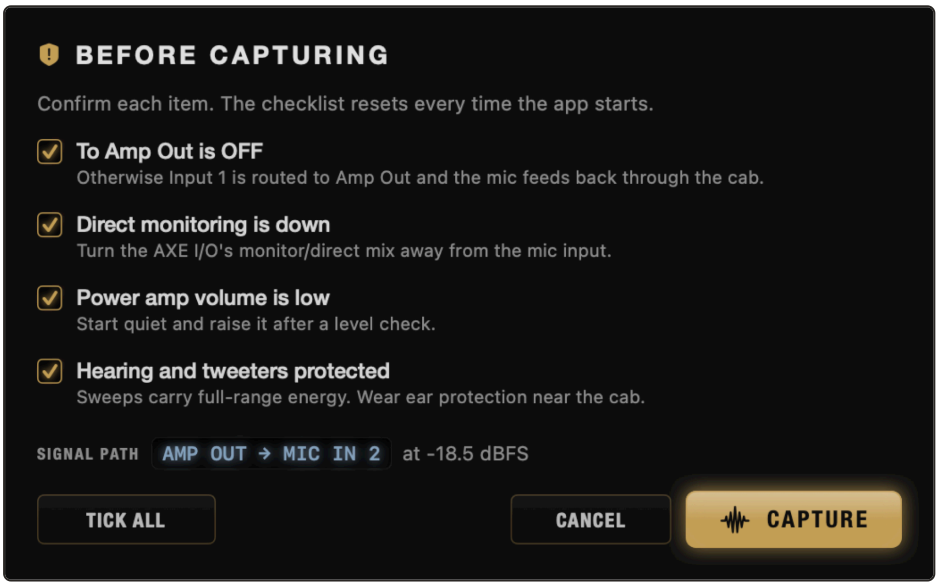
Start with the mic 2–3 cm from the grille, pointing at the edge of the speaker's dust cap. Moving towards the centre makes it brighter; moving out towards the cone edge makes it darker and warmer. Name each capture after the position (for example "V30 · SM57 cap edge") so you can compare them later.

3 Safety

> **WARNING** A sine sweep carries full-range energy, including high frequencies that can damage tweeters and your hearing. Always start quiet, wear ear protection near the cab, and never point a compression driver or tweeter at a sweep at high level.

IR Generator has several safeguards:

- **Level ceiling.** The output level is capped at −12 dBFS. Click the padlock to unlock the full range, and the slider turns red above −12 dBFS.
- **Soft starts.** Test signals fade in and out, and level changes are smoothed, so there are no clicks.
- **Instant stop.** `Esc` or the Stop button fades the output to silence immediately.
- **Checklist.** Before the first capture of each session the app asks you to confirm four things.



The pre-capture checklist. Clicking Capture opens it until every item is ticked. You can also tick the items in the **BEFORE CAPTURING** card in the left column.

4 A tour of the window



The main window, showing three captures overlaid in the Frequency view.

AREA	WHAT IT'S FOR
Top bar	The connected interface, its sample rate, and a **DEMO** badge in demo mode.
Left column	Capture set-up: interface and routing, test signal, output level, the safety checklist, and latency calibration.
Centre	The graphs, with tabs for **FREQUENCY**, **IMPULSE**, **RECORDING** and **SPECTROGRAM**. Each tab has its own controls in the second row. The legend and the selected capture's metrics are underneath.
Right column	The capture library, with **ALIGN** and **IMPORT** buttons. Underneath it, the **SELECTED IR** card: name, notes, details, and the **MIC SWAP**, **SAVE MASTER** and **EXPORT** buttons.
Bottom bar	Status and progress, output and input meters with peak readouts, the measured latency, and the **LEVEL CHECK**, **CAPTURE** and **EXPORT** buttons.

METRICS UNDER THE GRAPH

PNR	Peak-to-noise ratio: how far the IR's peak sits above the noise floor. 70 dB or more is a clean capture. Below 45 dB the app suggests raising the level or adding sweeps.
Delay	Time from the start of playback to the direct sound arriving: interface latency plus the mic's distance from the speaker.
Peak in	The highest input level during the capture. Aim for about −12 dBFS. At −0.1 dBFS or above, the app warns that the input clipped.
Length	Length of the master IR (0.5, 1 or 2 s).

5 Setting up a capture

Interface

DEVICE	Any duplex Core Audio interface. The AXE I/O is selected automatically when connected.
SEND TO	**AMP OUT** (outputs 5/6, instrument level, for an amp input) or **OUT 1/2**, **OUT 3/4** (line level, for a line-input power amp or the loopback test).
CHANNEL	Left, right or both outputs of the pair.
MIC INPUT	The AXE I/O input your mic is plugged into.

Test signal

SINE SWEEP	The most accurate method, and the default. Choose the **length** (2–10 s; longer gives more signal-to-noise) and the number of **sweeps** to average (×1 to ×8).
PINK / WHITE NOISE	A repeating noise burst, averaged over 4–32 periods. Useful in noisy rooms or as a cross-check. Pink is gentler on tweeters.
RANGE	The frequency range of the test signal, 20 Hz–20 kHz by default.
MASTER IR	How much of the IR to keep in the library: 500 ms, 1 s or 2 s. Exports are cut from this.
HIGH-PASS AT 20 HZ	Removes DC offset and sub-sonic rumble. Leave it on.

Output level and level check

The slider sets the level of the test signal leaving the AXE I/O. Click **LEVEL CHECK** (⌘L) to play looped pink noise or a 1 kHz tone, then:

1. Bring the power amp up from zero to a moderate playing volume.
2. Turn the AXE I/O's mic gain until the **IN** meter peaks around **–12 dBFS**. The peak readout holds for 1.5 s, then follows the level down.
3. Stop the level check. The sweep runs at the same level.

Latency & alignment (optional)

To measure the interface's round-trip latency, patch **Out 1** into **Input 2** (set to LINE), choose **OUT 1/2** and **IN 2**, and click **CALIBRATE**. With **KEEP MIC DISTANCE DELAY** ticked, IRs then keep the mic's time of flight.

> **TIP** USB timing varies by about ±1 ms between runs, which is too coarse for blending mics precisely. For most work leave **KEEP MIC DISTANCE DELAY** off, so each IR starts at the direct sound, and use **Align & Blend** to line up mics afterwards.

6 Capturing an IR

1. Wire up as in chapter 2 and set the AXE I/O as described there.
2. In the left column, choose **AMP OUT**, the channel, and the mic input.
3. Run a **LEVEL CHECK** and set the power amp and mic gain (chapter 5).
4. Click **CAPTURE** (⌘R). If the checklist isn't complete, tick it in the sheet that opens and click **CAPTURE** there.
5. Stay quiet while the sweep plays. The status bar shows progress and time remaining. Esc stops at any time.
6. The new capture appears at the top of the library and is selected. Give it a name and notes (mic, position, amp settings).
7. Check the result: a clean single spike in the **IMPULSE** view, a sensible curve in **FREQUENCY**, and a PNR of 70 dB or more.

> **LOOPBACK TEST** To check the app and interface on their own, patch Amp Out (or Out 1) straight into an input, choose that route and run a capture at a low level. The frequency response should be flat within about ±0.5 dB from 20 Hz to 20 kHz, with a single clean spike in the impulse view.

If a capture looks wrong

- **Noisy or low PNR:** raise the level, use a longer sweep, or average more sweeps.
- **Clipping warning:** turn the mic gain down and capture again.
- **Strange start to the impulse** with **KEEP MIC DISTANCE DELAY** on: the latency moved since calibration. The app falls back to trimming at the direct sound and says so. Right-click the capture and choose **REPROCESS FROM RECORDING** after changing settings.

7 Reading the graphs

The graphs show the selected capture plus any others whose **eye** button is on in the library, each in its own colour. Hover over a graph for an exact readout.

Frequency

The magnitude response on a logarithmic scale from 20 Hz to 20 kHz. Each curve is scaled so its loudest point between 40 Hz and 12 kHz sits at 0 dB, so captures at different levels are easy to compare, and the dB scale fits itself to the curves. **SMOOTH** averages over 1/12, 1/6 or 1/3 of an octave; **HIGH CUT** previews the export filter (chapter 9); **LENGTH** shows the response as it will be after cutting to an export length. Short IRs lose low-end detail, and this shows how much.

A typical guitar cab rises from the low end to a peak around 80–120 Hz, has presence peaks between 2 and 5 kHz, and falls steeply above 5–6 kHz. The wiggles in the upper mids are the speaker's cone breakup, part of its character.

Impulse



Impulse view, waveform, first 50 ms. The dashed lines show where 1024- and 2048-sample exports end.

WAVEFORM shows the IR itself; **ENERGY (DB)** shows its decay on a log scale, which makes the noise floor and late resonances visible. The span buttons zoom from 5 ms to the whole IR.

Recording



Recording view: the raw mic signal for the capture, with the clip level marked.

Use this to check the level was sensible and nothing unexpected (a door slam, a car) happened during the sweep.

Spectrogram



Spectrogram of the recording: time across, frequency up, brightness for level.

The bright rising line is the sweep. Fainter lines above it are harmonic distortion from the power amp or speaker; the deconvolution moves these ahead of the IR and they're trimmed away. Horizontal lines that run through the whole recording are hum or background noise.

8 The capture library

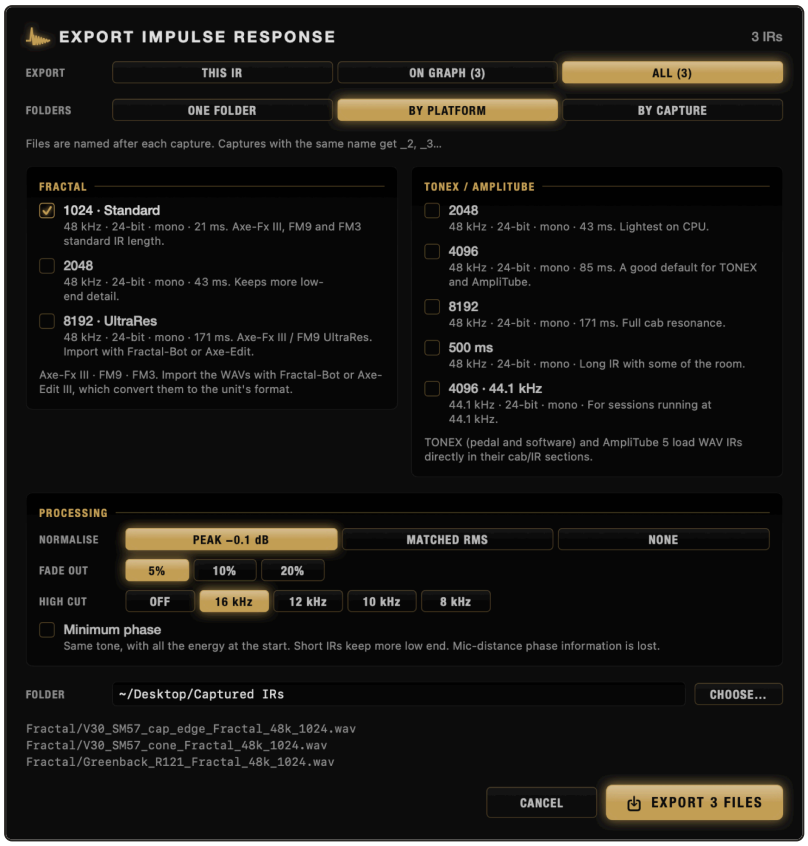
Every capture is saved automatically, with a 32-bit float master IR, the raw recording and its settings. The library survives relaunches, and you can re-export any capture with different settings later.

- **Select** a capture to show it in gold and fill the **SELECTED IR** card. Edit its name and notes there.
- **Eye** button: add the capture to the graphs to compare it with the selected one.
- **IMPORT** (`⌘I`): load WAV IRs from elsewhere (commercial IRs, for example) to compare with your own.
- Labels show **IMPORTED** IRs and **PROCESSED** ones (aligned, blended or mic-swapped). Processed IRs remember the capture they came from.

RIGHT-CLICK MENU

EXPORT…	Open the export sheet for this capture.
SAVE MASTER…	Save the full-length 32-bit float IR as a WAV.
REPROCESS FROM RECORDING	Rebuild the IR from the raw recording with the current settings (high-pass, keep delay, master length).
SAVE RAW RECORDING…	Save the mic recording, for example to process in other software.
SHOW ORIGINAL	For processed IRs, select the capture it was made from.
DELETE	Remove the capture and its files.

9 Exporting for your modellers



The Export sheet (⌘E), here exporting the whole library by platform. Tick every format you need; one click writes them all.

PRESET	RATE	LENGTH	USE
Fractal Standard	48 kHz	1024 (21 ms)	Axe-Fx III, FM9, FM3 standard IR slots
Fractal 2048	48 kHz	2048 (43 ms)	More low-end detail
Fractal UltraRes	48 kHz	8192 (171 ms)	Axe-Fx III / FM9 UltraRes
TONEX / AmpliTube	48 kHz	2048 · 4096 · 8192 · 500 ms	4096 is a good default
TONEX / AmpliTube	44.1 kHz	4096	Sessions at 44.1 kHz

All exports are mono 24-bit WAV, named `<name>_<Fractal|TONEX>_<rate>_<length>.wav`.

EXPORTING EVERYTHING AT ONCE

The **EXPORT** row at the top of the sheet sets what's exported: **THIS IR**, **ON GRAPH** (every IR whose eye is on) or **ALL**. Open it straight in **ALL** with ⇧⌘E, **EXPORT ALL…** in a capture's right-click menu, or the export-all button in the library header. A batch writes every ticked preset for every IR, using the same processing options. **FOLDERS** chooses **ONE FOLDER**, **BY PLATFORM** (`Fractal/` and `TONEX/`, ready to drag into each editor) or **BY CAPTURE**. Files are named after each capture; captures that share a name get _2, _3… so nothing in the batch overwrites anything else. Imported IRs are left out unless you tick **INCLUDE IMPORTED IRS**.

PROCESSING OPTIONS

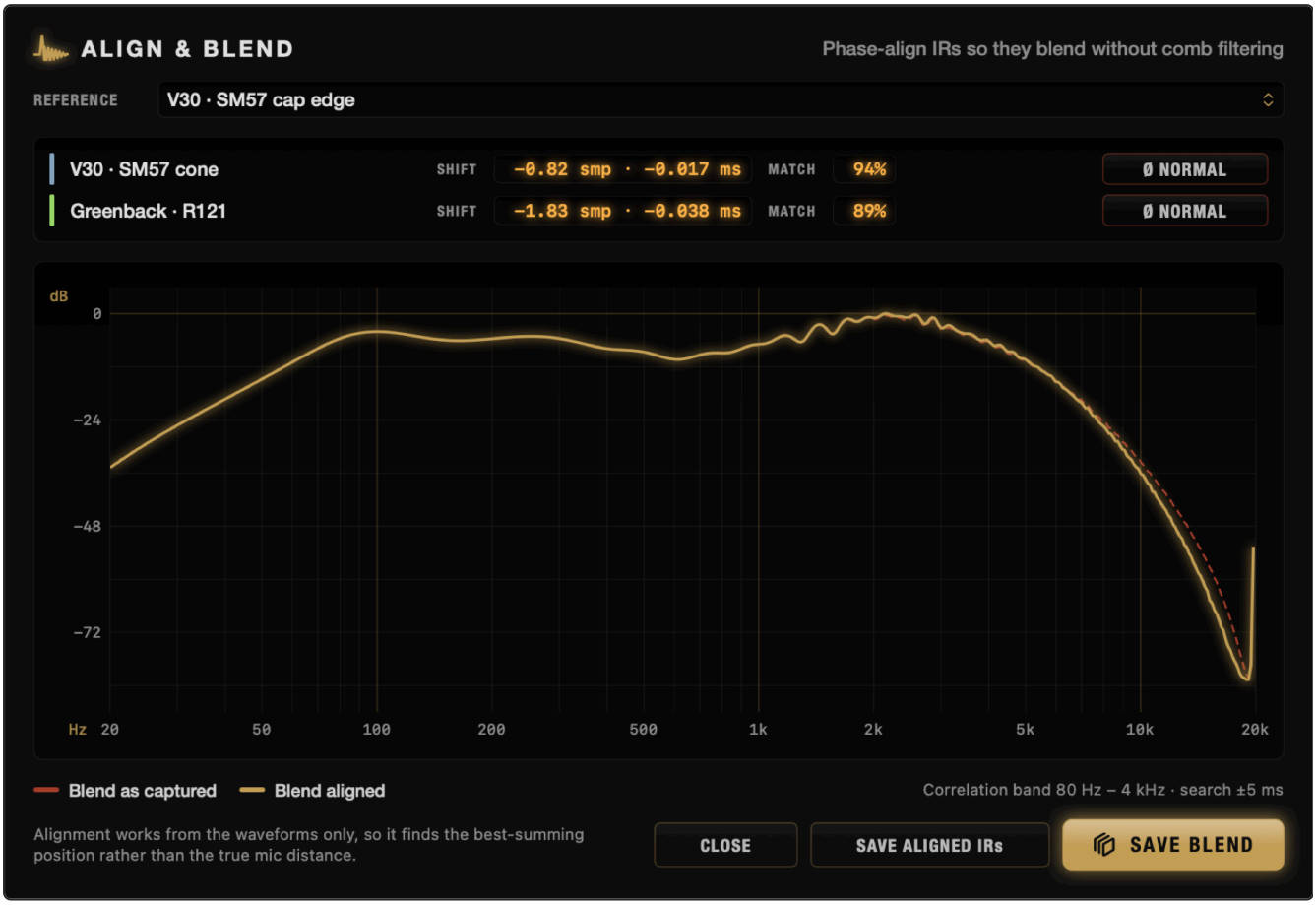
NORMALISE	**Peak –0.1 dB** (the loudest sample just below full scale, the usual choice), **Matched RMS** (every IR at the same loudness, good for A/B comparison in a modeller), or **None**.
FADE OUT	Length of the half-Hann fade at the end of each file (5, 10 or 20%), so cutting the IR short doesn't click.
HIGH CUT	A 24 dB/octave low-pass. See below.
MINIMUM PHASE	Moves all the energy to the start of the IR without changing its tone. Short IRs keep more low end, but the timing between mics is lost, so don't use it on IRs you'll blend.

SHOULD I USE THE HIGH CUT?

Usually, yes, set gently. A guitar speaker puts out very little above about 6–8 kHz. What the capture shows up there is mostly mic hiss and noise at the edge of the sweep, which a high-gain amp turns into fizz. The filter is flat below its cutoff, so **10–12 kHz** cleans the top without changing the tone. **8 kHz** and below starts to dull the pick attack. The master IR in the library is never changed, so try a few settings and compare by ear.

10 Align & Blend

Blending two mics (for example an SM57 and a ribbon) is a classic way to build a fuller cab sound. If the two IRs are even a fraction of a millisecond apart, though, they cancel at some frequencies and sound thin and phasey (comb filtering). Align & Blend lines them up first.



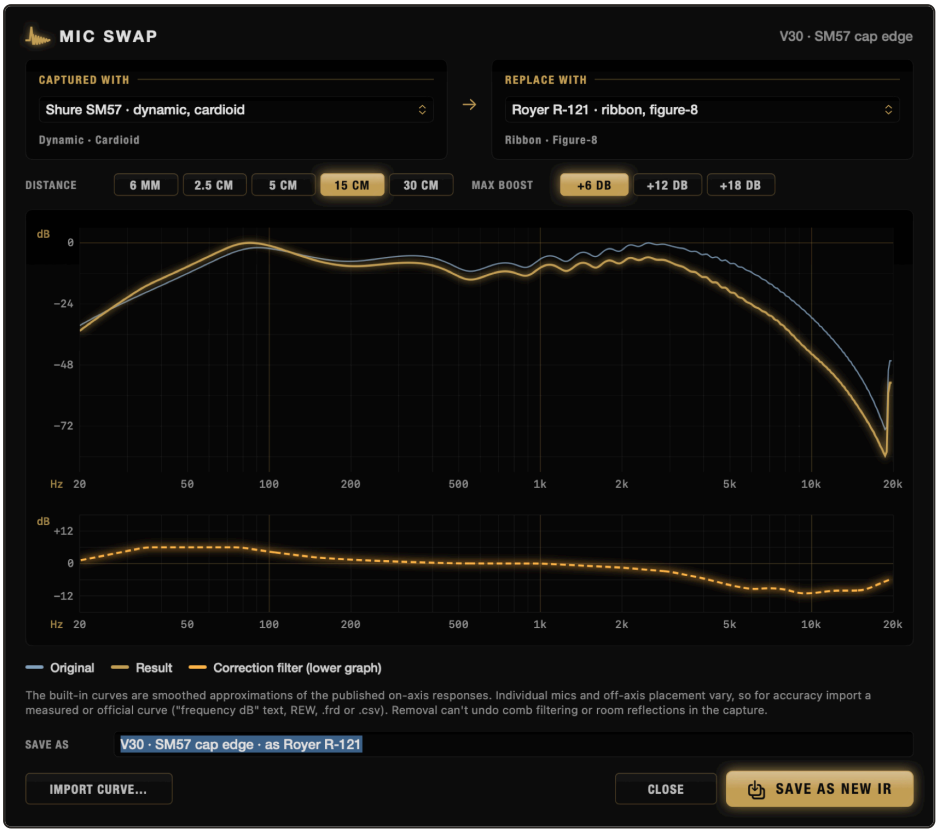
Align & Blend (⌘B). Gold: the blend after alignment. Red dashed: the blend as captured.

1. Select one capture and turn on the **eye** of each IR you want to blend with it, then click **ALIGN** in the library header.
2. Choose the **REFERENCE**. The others are shifted to match it. **SHIFT** shows how far, to a hundredth of a sample, and **MATCH** how similar the two are.
3. Check polarity. An IR that's upside down (a ribbon's back side, for example) is detected and flipped; the **Ø** button overrides it.
4. With two IRs, set the **MIX**. The graph shows the blend before and after alignment.
5. **SAVE BLEND** adds the blend to the library; **SAVE ALIGNED IRS** saves the shifted IRs individually, for blending in a modeller with two IR slots.

> **NOTE** Alignment works from the waveforms, between 80 Hz and 4 kHz, so it finds the position where the mics sum best rather than their physical distance. That's what you want for a blend.

11 Mic Swap

Every mic colours the sound. Mic Swap divides out the capture mic's frequency response to leave the speaker on its own, or replaces it with another mic's response.



Mic Swap (⌘M): an SM57 capture re-voiced as a Royer R-121 at 15 cm. The lower graph shows the correction filter.

1. Select a capture and click **MIC SWAP…**.
2. **CAPTURED WITH**: the mic you used. **REPLACE WITH**: the target mic, or **Flat** for a speaker-only IR.
3. Set the **DISTANCE** from the grille. Directional mics boost bass when close (the proximity effect), and the app models this for both mics.
4. Set the **MAX BOOST**. Large boosts also raise noise; +6 dB is a safe start.
5. Compare the result with the original, then **SAVE AS NEW IR**.

Built-in curves: SM57, MD 421, e906, i5, RE20, SM7B (dynamic); R-121, M 160 (ribbon); C414, U 87 (condenser); and flat. They're smoothed approximations of published on-axis responses, not manufacturer data. For better accuracy, click **IMPORT CURVE…** and load a measured or official response as a text, REW, `.frd` or `.csv` file of frequency and dB pairs.

> **LIMITS** A swap is an estimate. It corrects the on-axis tone, but off-axis colouration, comb filtering and room reflections in the original capture stay in. It's no substitute for capturing with the real mic, but it's a fast way to explore.

12 Loading IRs into TONEX, AmpliTube and Fractal

TONEX and AmpliTube

TONEX (software and pedal, via the TONEX editor) and AmpliTube 5 load standard WAV IRs directly in their cab / IR sections. Use the **TONEX / AmpliTube** presets, 4096 samples at 48 kHz unless you have a reason to change it. Choose the 44.1 kHz version only if your session or host runs at 44.1 kHz.

Fractal Axe-Fx III, FM9 and FM3

Import the WAVs with **Fractal-Bot** or **Axe-Edit III**, which convert them to the unit's own format and store them in a user cab slot. Export **1024** for the standard slots, **2048** where supported, and **8192 UltraRes** for UltraRes slots on the Axe-Fx III and FM9.

> **TIP** When auditioning several IRs in a modeller, export them with **Matched RMS** normalisation so louder isn't mistaken for better.

13 Troubleshooting

PROBLEM	WHAT TO TRY
No audio device, or the AXE I/O isn't listed	Check the USB connection and that macOS sees the AXE I/O in Audio MIDI Setup. The device list refreshes on its own.
Input meter doesn't move	Allow microphone access in System Settings › Privacy & Security › Microphone. Check the mic input selection and the AXE I/O gain, and phantom power for condensers.
A batch export is missing an IR	Imported IRs are only included in **ALL** when **INCLUDE IMPORTED IRS** is ticked. **ON GRAPH** only covers IRs whose eye is on.
Capture button does nothing	The safety checklist isn't complete: tick the items in the sheet or in the **BEFORE CAPTURING** card.
Howling or feedback	Press `Esc`. Turn **To Amp Out** off and direct monitoring down on the AXE I/O.
Low PNR, hissy IR	Raise the output level (and power-amp volume), use a longer sweep or more sweeps, and reduce room noise. A 10–12 kHz high cut on export also helps.
Clipping warning	Turn the mic gain down so the input peaks around −12 dBFS.
The impulse looks cut off at the start	Turn off **KEEP MIC DISTANCE DELAY** (or recalibrate), then use **REPROCESS FROM RECORDING**.
Blend sounds thin or phasey	Use Align & Blend, and check polarity. Don't export IRs for blending as minimum phase.

14 Reference

KEYBOARD SHORTCUTS

`⌘R`	Capture
`⌘L`	Level check
`Esc`	Stop
`⌘E`	Export the selected IR
`⇧⌘E`	Export all captures
`⌘B`	Align & Blend
`⌘M`	Mic Swap
`⌘I`	Import WAV IRs

WHERE FILES ARE KEPT

`~/Library/Application Support/IR Generator/`

- `Captures/<id>/meta.json`: settings, name, notes
- `Captures/<id>/ir.wav`: master IR, 32-bit float
- `Captures/<id>/recording.wav`: raw mic recording
- `Mics/`: imported mic curves

Exports go to `~/Documents/Impulse Responses` unless you choose another folder.

HOW IT WORKS

Sweep	Exponential sine sweep (Farina): $x(t) = \sin(2\pi f_1 L (e^{t/L} - 1))$, $L = T / \ln(f_2/f_1)$, with faded ends and averaging.
Noise	Band-limited periodic pink or white noise, one 2^{16}-sample period built in the frequency domain, synchronously averaged and deconvolved circularly.
Deconvolution	$H = Y \cdot X^* / (\lvert X\rvert^2 + \varepsilon(f))$, double-precision FFT. ε is tiny in band and rises over a third of an octave outside it, so out-of-band noise isn't amplified.
Trimming	Onset at the first sample within –20 dB of the peak; 0.2 ms kept before it and faded in. Optional 20 Hz 2nd-order Butterworth high-pass.
Audio engine	One Core Audio IOProc, written in C, plays and records in the same callback: the recording is sample-aligned with playback and the audio thread never runs Swift.
Export	Optional minimum phase (real cepstrum), Kaiser windowed-sinc resampling with gain preserved, optional 4th-order Butterworth high cut, half-Hann fade, normalisation.
Alignment	Cross-correlation band-limited to 80 Hz–4 kHz over ±5 ms, parabolic sub-sample peak, polarity from the peak's sign, fractional shift by FFT phase ramp.
Mic Swap	Minimum-phase correction FIR from the dB ratio of the two mic curves, with each pattern's proximity effect at the set distance, limited boost and cut, faded to 0 dB below 35 Hz and above 16 kHz.

LICENCE AND TRADEMARKS

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